PREOPERATIVE DIAGNOSIS: Displaced, unstable right intertrochanteric hip fracture (AO/OTA 31-A2) in an 82-year-old female following a ground-level fall.

POSTOPERATIVE DIAGNOSIS: Displaced right intertrochanteric femur fracture.

PROCEDURE PERFORMED: Closed reduction and internal fixation with a Stryker Gamma3 Long Intramedullary Cephalomedullary Nail (10 mm x 340 mm x 125-degree lag screw angle).

PROCEDURAL NARRATIVE: Under spinal anesthesia and intravenous sedation, the patient was positioned supine on a fracture table. The right lower extremity was placed into a traction boot with gentle longitudinal traction, 15 degrees of internal rotation, and slight adduction. Fluoroscopic C-arm confirmation in both AP and lateral planes demonstrated excellent reduction of the head-neck fragment with restoration of the shenton line and neutral neck-shaft angle.

The right hip, lateral thigh, and iliac crest were prepped and draped in standard orthopedic sterile fashion. A 4 cm longitudinal incision was placed 2 cm proximal to the tip of the greater trochanter. The gluteus medius fascia was split longitudinally in line with its fibers. Under biplanar fluoroscopic guidance, a cannulated curved awl was introduced into the trochanteric fossa entry point at the junction of the anterior one-third and posterior two-thirds of the trochanter. A ball-tipped 3.0 mm guidewire was passed through the entry site, across the fracture zone, and advanced into the distal femoral medullary canal to the level of the physeal scar.

Sequential flexible intramedullary reaming was carried out from 9.0 mm to 11.5 mm in 0.5 mm increments. A 10 mm x 340 mm Gamma3 titanium nail with 125-degree proximal angle was mounted onto the carbon-fiber insertion guide and inserted gently over the guidewire into the femoral shaft without forceful hammering. Through the targeting arm, a guide sleeve was passed to the lateral cortex, and a central guide pin was advanced through the femoral neck into the subchondral bone of the femoral head. A 10.5 mm x 95 mm lag screw was placed with a tip-apex distance (TAD) calculated at 14 mm on combined AP and lateral views. Distal dynamic locking was achieved through the targeting jig with a 5.0 mm cortical locking bolt. Final fluoroscopic views confirmed stable fixation, anatomical alignment, and rigid compression across the fracture fragments. The wounds were copiously irrigated with sterile saline, and the fascia and skin closed in layers with 0 Vicryl and surgical staples.